Marketing Analytics

Chapter 5 · Learning from Observational Data

Nils Wlömert
Vienna University of Economics and Business (WU)

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Part 1 · Observational data I

Before we start

Paper quiz on Chapter 4 *(10 min)*

- Individual, closed book
- Counts toward participation

Assignment 2 is running

- Deadline: Group A Sun, Nov 8 · Group B Wed, Nov 11
- Debrief at the start of next session

Today’s session

Observational data I *(today)*

5.1 Smart Mix without randomization
5.2 From correlation to a line
5.3 Reading a regression
5.4 Adjusting for confounders
5.5 Categorical predictors
5.6 Prediction vs. explanation

Observational data II *(next session)*

5.7 Functional form and elasticities
5.8 Logistic regression
5.9 Diagnostics that change decisions
5.10 Difference-in-differences

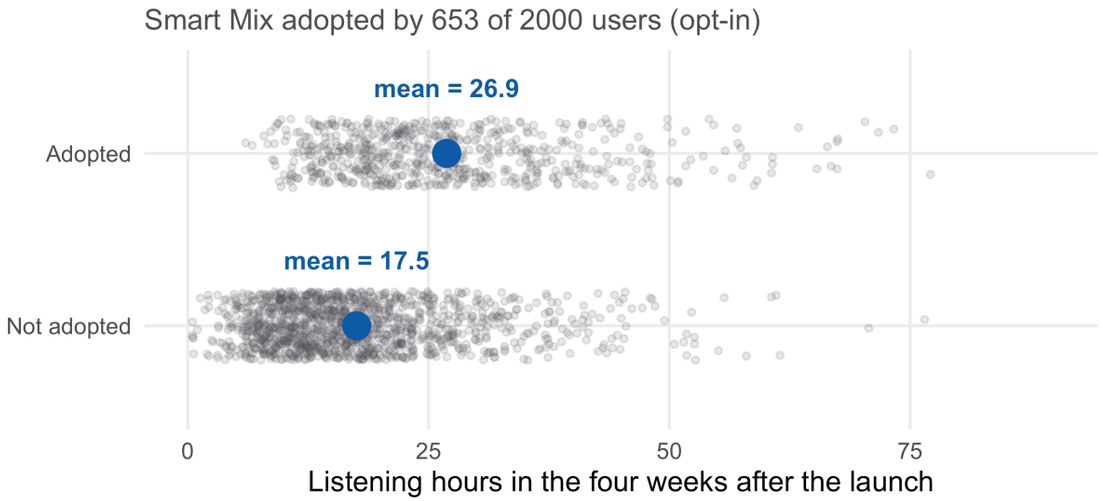
> **Driving question:** *What if we cannot randomize?*

5.1 Smart Mix without randomization

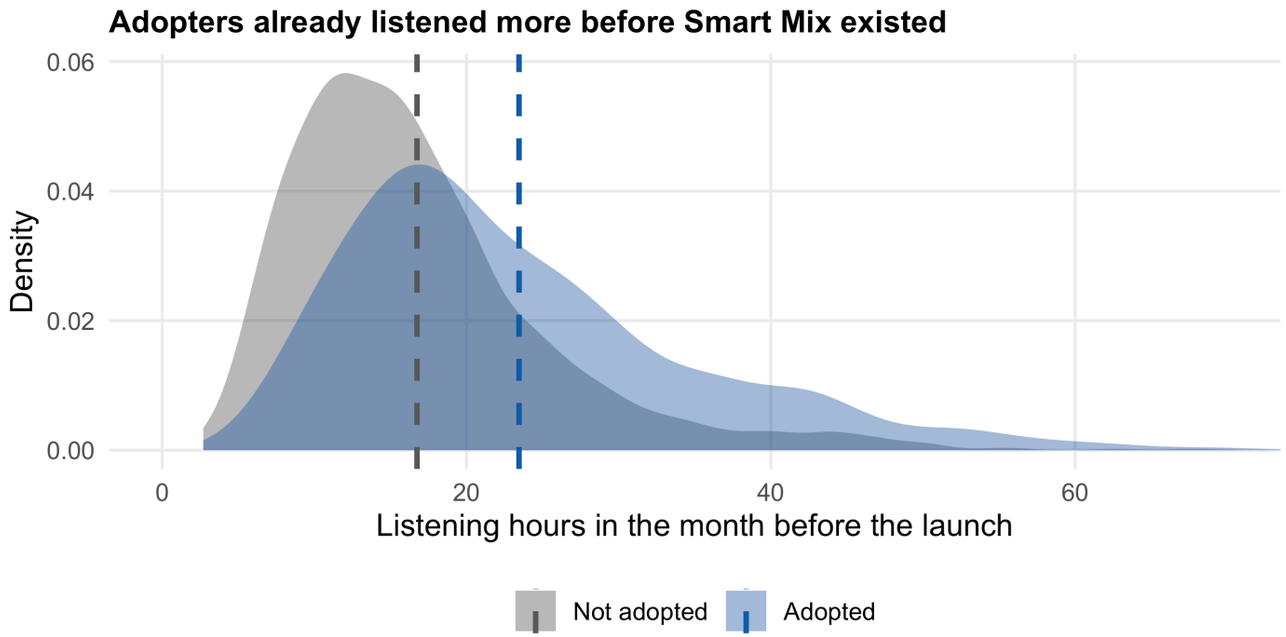
A different history

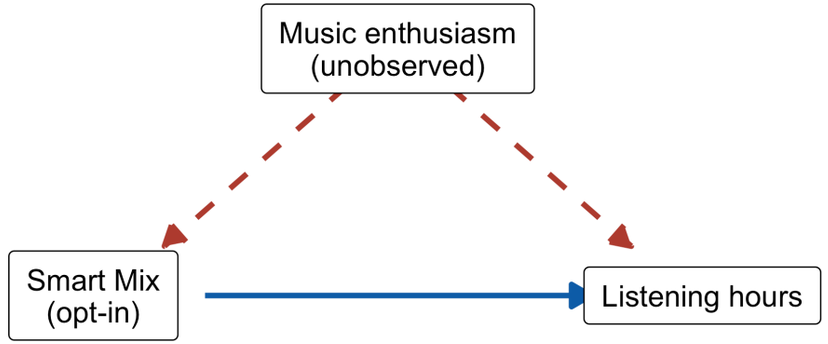
- Chapter 4: A/B test, 400 users → **+3.2 hours**
- Now: no experiment. Smart Mix launched as an **opt-in** feature
- Four weeks later: data of **2,000 users**, 653 adopters

> The dashboard says: adopters listen **9.3 hours more** — three times the experimental effect. Would you report this number to management?



Adopters were different before Smart Mix existed

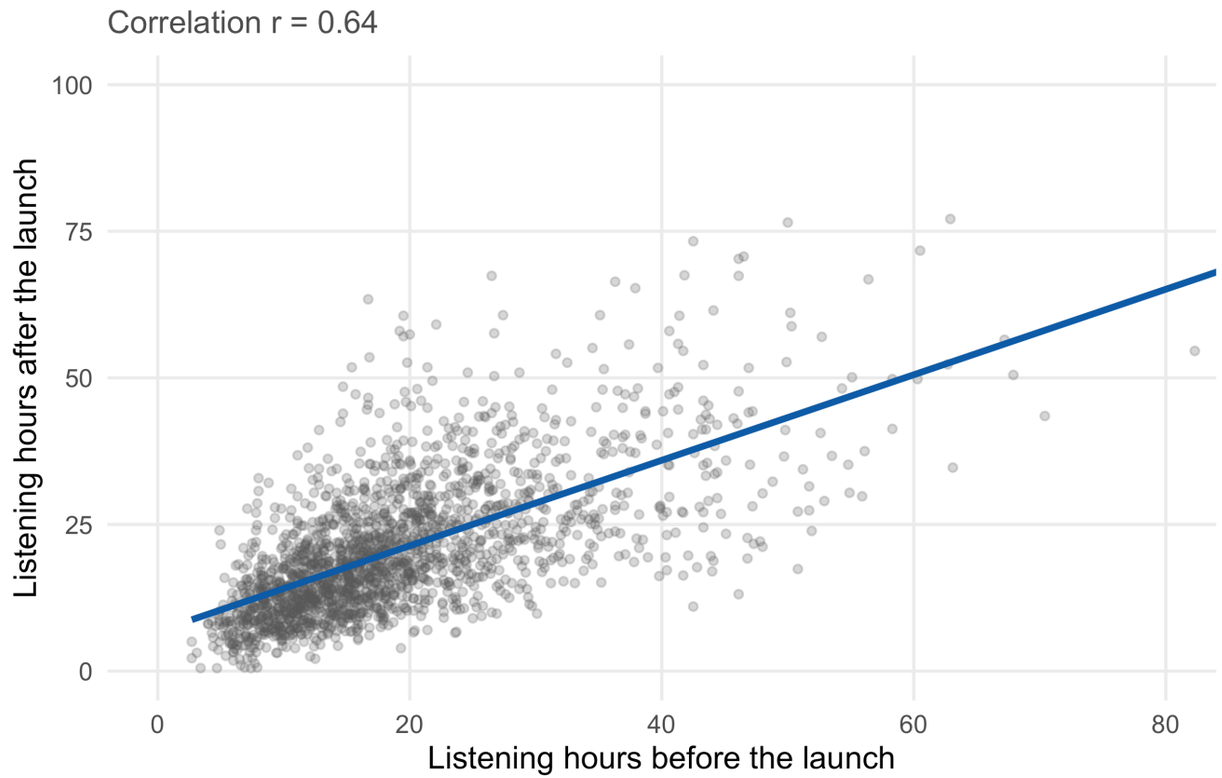




Same costume, new villain: **music enthusiasm** drives adoption *and* listening. Without a coin flip, we have one tool left: **adjustment**.

5.2 From correlation to a line

Two variables, one line



$$\text{hours} = \beta_0 + \beta_1 \cdot \text{prior hours} + \varepsilon$$

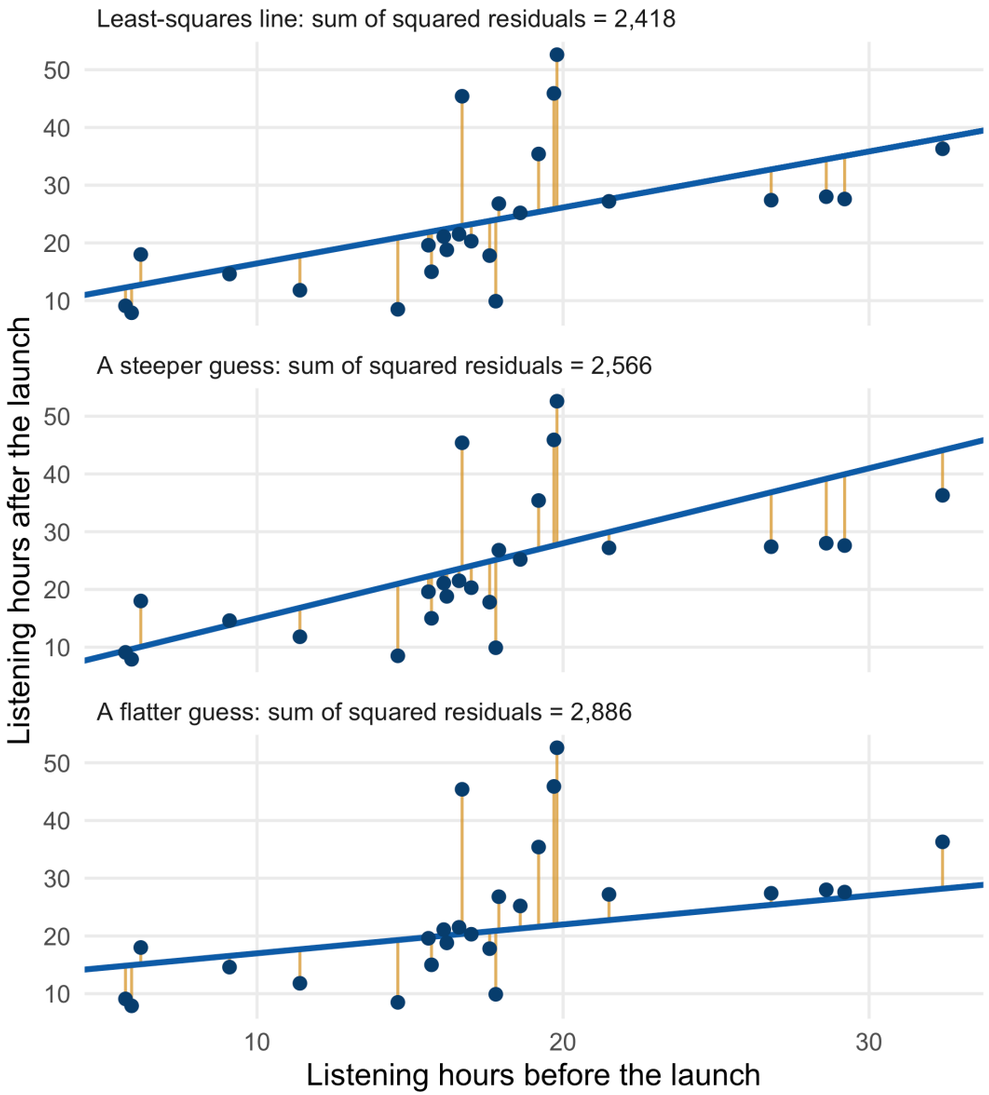
- β_0 **intercept** · β_1 **slope** · ε **residual**
- Correlation: *that* they move together · regression: *how much*

```
1  coef(lm(hours ~ prior_hours, data = smartmix_obs))
```

```
(Intercept) prior_hours
     6.7849      0.7291
```

> Each extra hour before the launch **goes with** 0.73 hours after it — *goes with*, not *causes*.

Which line? Least squares



- Residual = observed − predicted (orange segments)
- The regression line **minimizes the sum of squared residuals**
- Squaring: big misses count a lot; above and below the line count alike
- `lm()` = “linear model”: does exactly this

Live demo: fit the line yourself — course book, §5.2

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-line

Screenshot placeholder: the line-fitting widget
`img/ch05/widget_line.png`

5.3 Reading a regression

The full output

```r
model_naive <- lm(hours ~ smart_mix, data = smartmix_obs)
summary(model_naive)
```

```
Call:
lm(formula = hours ~ smart_mix, data = smartmix_obs)

Residuals:
   Min     1Q Median     3Q    Max
-20.88  -7.64  -2.06   4.76 101.22

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   17.541      0.300    58.5   <2e-16 ***
smart_mix      9.341      0.525    17.8   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11 on 1998 degrees of freedom
Multiple R-squared:  0.137, Adjusted R-squared:  0.136
F-statistic:  316 on 1 and 1998 DF,  p-value: <2e-16
```

Line by line

Output	Meaning
`Estimate`	Intercept = mean of non-adopters (18.4); `smart_mix` = difference adopters − non-adopters (9.3)
`Std. Error`	Uncertainty of each coefficient (Chapter 3)
`t value`, `Pr(>|t|)`	*t*-test of each coefficient against zero (Chapter 4): signal / noise
`Residual standard error`	Typical distance of a user from the line, in hours
`Multiple R-squared`	Share of the variation in hours explained by the model (14%)
`F-statistic`	The whole model against “no predictors”

> With one 0/1 predictor, the regression **is** the comparison of two means — and its *t*-test is the two-sample *t*-test. `confint(model_naive)` gives the confidence intervals.

5.4 Adjusting for confounders

Holding prior listening constant

```r
model_prior <- lm(hours ~ smart_mix + prior_hours, data = smartmix_obs)
coef_table(model_prior)
```

```
            Estimate Std. Error t value p-value
(Intercept)    6.451      0.412   15.65 < 0.001
smart_mix      4.901      0.443   11.08 < 0.001
prior_hours    0.662      0.020   33.17 < 0.001
```

> The Smart Mix coefficient drops from **9.3** to **4.9**: now the difference between adopters and non-adopters **with the same prior listening**. Part of the naive gap belonged to prior listening — and is now attributed to it.

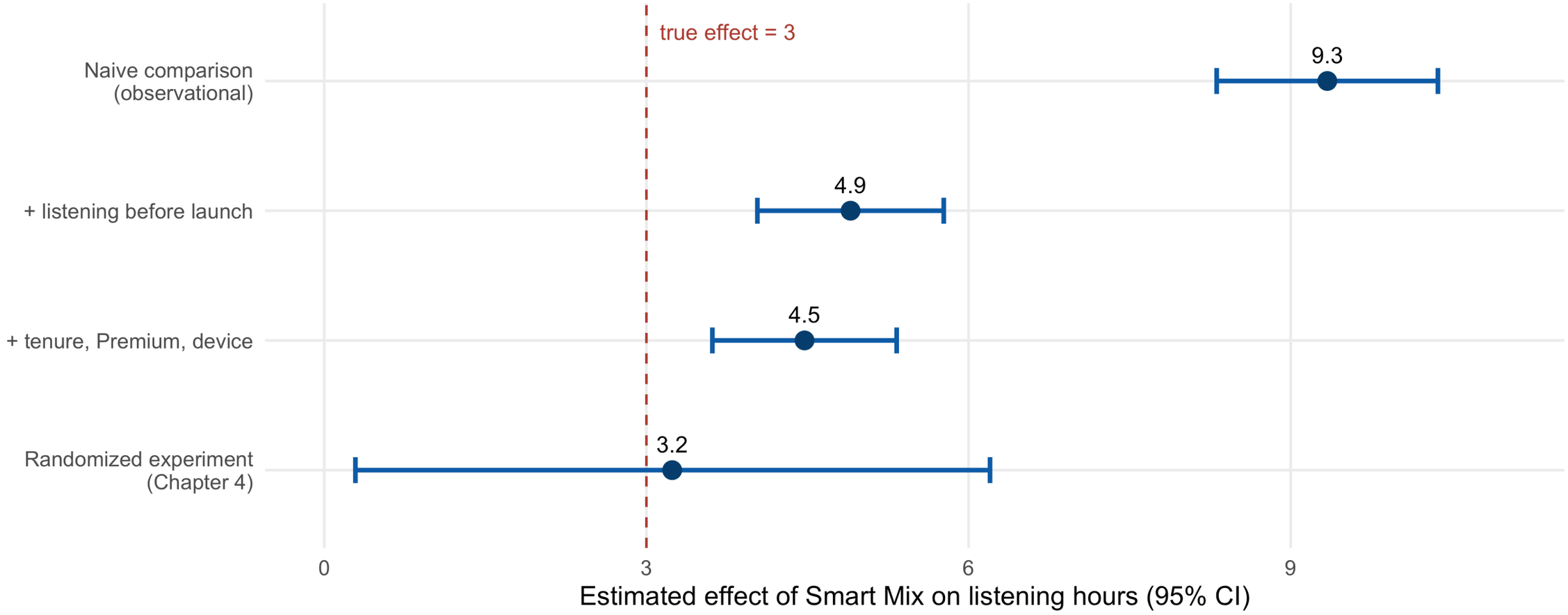
More variables, same logic

```
model_full <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device, data = smartmix_obs)
coef_table(model_full)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	6.255	0.540	11.58	< 0.001
smart_mix	4.472	0.437	10.22	< 0.001
prior_hours	0.613	0.021	29.69	< 0.001
tenure	0.038	0.011	3.34	< 0.001
premium	2.924	0.426	6.87	< 0.001
deviceAndroid	−1.475	0.422	−3.49	< 0.001
deviceDesktop	−2.488	0.576	−4.32	< 0.001

> Each coefficient: the association with the outcome **holding all other variables in the model constant**. Tenure, Premium and device change little: ≈ **4.5 hours**.

Naive, adjusted — and the truth



Adjustment removed ~75% of the bias — **not all of it**. We adjusted for a *proxy* of enthusiasm, not for enthusiasm. Regression removes exactly the confounding your variables capture.

Live demo: how good must a proxy be?

Course book, §5.4 — set the **quality of the proxy** and the **strength of confounding**.

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-adjust

Screenshot placeholder: the proxy widget
`img/ch05/widget_proxy.png`

> With a 50% proxy, how much of the bias is left? And what happens at 0%?

Two warnings

Adjusting for the wrong variable makes it worse. A **mediator** (on the causal path) absorbs part of the effect you want. Only **confounders** — determined *before* the treatment, affecting treatment *and* outcome — belong in the model.

"Enough" is untestable. The assumption of no unobserved confounders cannot be checked in the data. Report *what* you adjusted for — and what you plausibly could not.

Your turn (5 min)

Add **age** to the full model. Does the Smart Mix coefficient change? Why (not)?

```
model_age <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device + age,
                data = smartmix_obs)
coef_table(model_age)
```

Then: which variable in the data set should you **never** adjust for when estimating the effect of Smart Mix on listening — and why?

5.5 Categorical predictors

Dummies and the reference category

```
coef(model_full)[c("premium", "deviceAndroid", "deviceDesktop")]
```

```
premium deviceAndroid deviceDesktop
  2.924        −1.475        −2.488
```

- **0/1 variable** (`premium`): coefficient = difference between the two groups, other variables held constant
- **Several categories** (`device`): R picks a **reference** (first level: iOS) and creates one **dummy** per other category
- Each dummy = difference to the **reference**

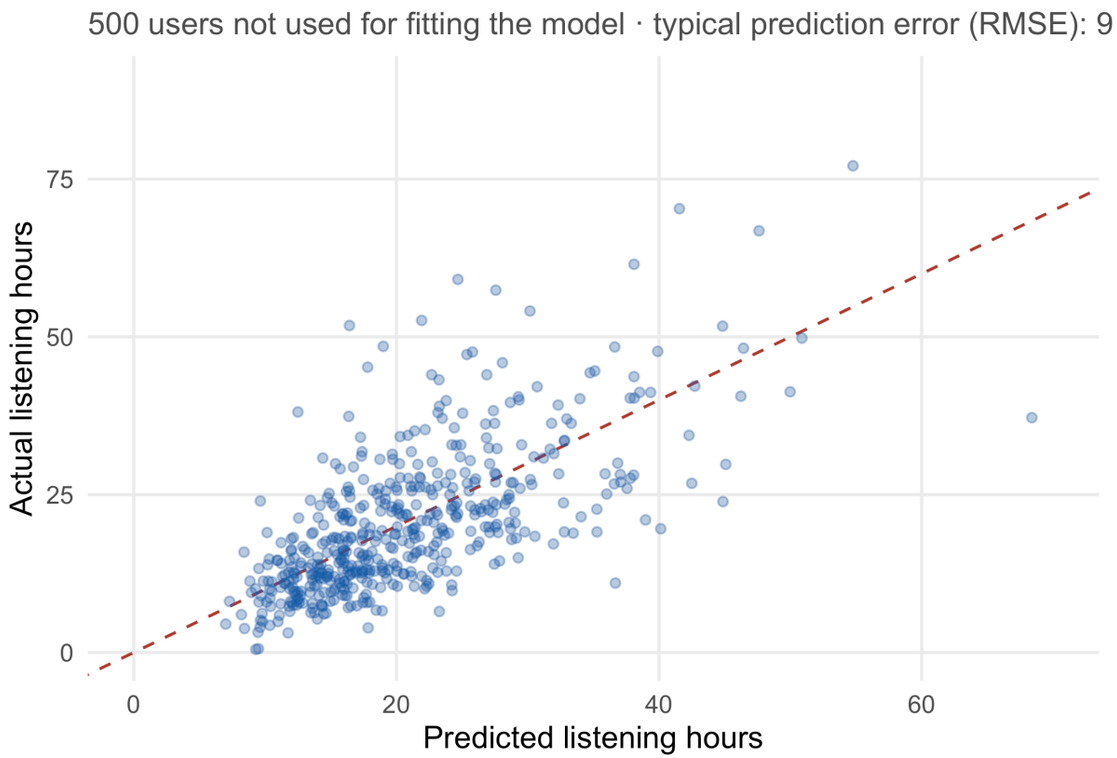
> Android users: **1.5 hours less** than iOS users; Desktop: **2.5 hours less** — holding everything else constant. *p*-values compare each category with the reference; “does device matter at all?” → `anova(model_without, model_full)` (the ANOVA of Chapter 4).

5.6 Prediction vs. explanation

Predicting listening time for new users

```r
set.seed(42)
train_ids <- sample(nrow(smartmix_obs), 1500)
train <- smartmix_obs[train_ids, ]
test  <- smartmix_obs[-train_ids, ]
model_pred <- lm(hours ~ prior_hours + tenure + premium +
                   device + smart_mix, data = train)
test$predicted <- predict(model_pred, newdata = test)
sqrt(mean((test$hours - test$predicted)^2))   # RMSE
```

```
[1] 9.425
```



For **prediction**, it does not matter *why* a variable is associated with the outcome — only how well the predictions hold up on **data the model has not seen**.

Two questions, two rules

Explanation (causal question): include confounders, exclude mediators, state what is unobserved. The **coefficient** matters; R^2 is secondary.

Prediction: include whatever predicts; judge by **out-of-sample errors**. Coefficients are not interpreted; the **predictions** matter.

A model can be a **good predictor and a bad explainer** at the same time. Where in your company would each mistake — reading causal effects off a prediction model, or judging a causal model by its R^2 — cost money?

Learning check (Part I)

1. Slope of hours on prior hours = 0.73. Interpret — and why is it not causal?
2. Adding a variable can increase, decrease, or leave the Smart Mix coefficient unchanged. When does each happen?
3. `planFamily` = 4.2 with reference “Free”. Interpret. How do you test whether the plan matters at all?
4. R^2 = 0.71 for predicting next month’s spending with 25 predictors on 300 customers. What do you ask?

Next session

Observational data II — beyond the straight line, and a quasi-experiment

- Functional form: when a line is wrong — **elasticities**
- Yes/no outcomes: logistic regression
- The three diagnostics that change decisions
- **Difference-in-differences** with real playlist data

> Before next session: work through §5.1–5.6 on the course site. **Debrief of Assignment 2** at the start of the session.

Part 2 · Observational data II

Before we start

Debrief Assignment 2 — analyzing experiments *(10 min)*

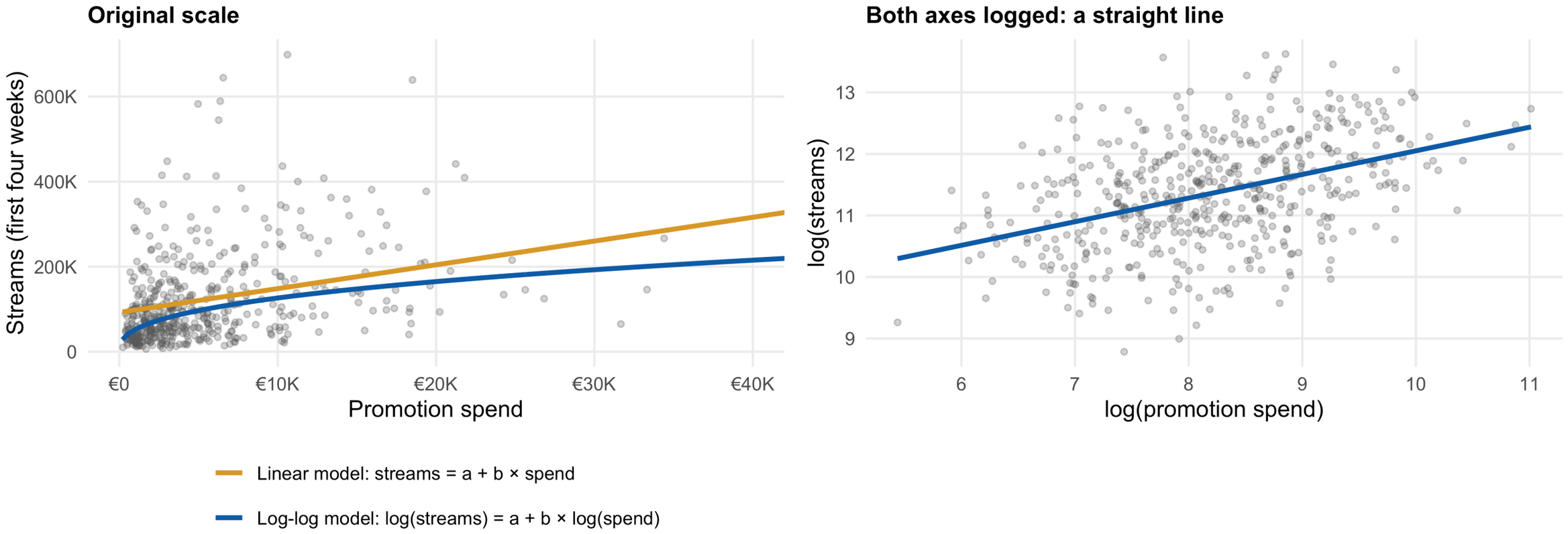
- Three short paper questions about *your* results
- Closed book, no laptop

Paper quiz on Chapter 5, Part I *(§5.1–5.6, 8 min)*

- Individual, closed book
- Counts toward participation

5.7 Functional form: when the line is wrong

Diminishing returns



500 promotion campaigns: the first euros buy many streams, later euros fewer. A straight line cannot bend — on **log scales**, the cloud becomes a line again.

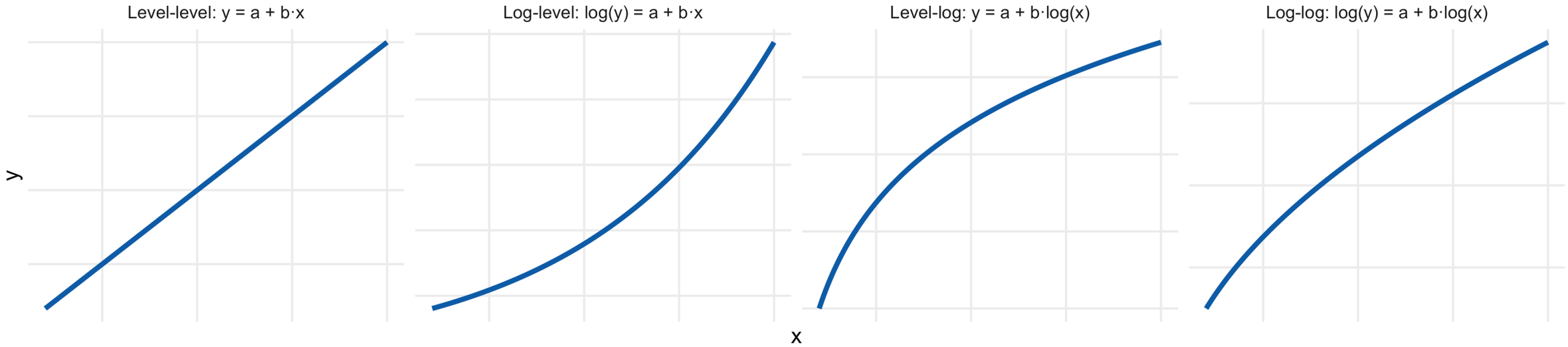
The log-log model: elasticities

```
1 model_loglog <- lm(log(streams) ~ log(spend) + log(followers) + placements + genre, data = campaigns)
2 coef_table(model_loglog)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	2.881	0.235	12.28	< 0.001
log(spend)	0.331	0.021	16.00	< 0.001
log(followers)	0.507	0.016	32.28	< 0.001
placements	0.120	0.012	10.07	< 0.001
genreHipHop/Rap	0.199	0.048	4.19	< 0.001
genreRock	−0.256	0.060	−4.27	< 0.001
genreElectro/Dance	0.103	0.059	1.73	0.085

> `log(spend)` = **0.33**: a 1% increase in spend goes with 0.33% more streams; +10% spend → **+3.3% streams** — the same *percentage* at every budget. Elasticities have no units and are **comparable**: followers (0.51) matter more than the budget.

Four ways to combine levels and logs



Model	Equation	Reading β_1
Level-level	$y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \beta_1$ units of y
Log-level	$\log y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \approx 100\beta_1\%$ of y (exact: $100(e^{\beta_1} - 1)\%$)
Level-log	$y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1/100$ units of y
Log-log	$\log y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1\%$ of y (**elasticity**)

Your turn (5 min)

From the log-log model:

1. `genreHipHop/Rap` = 0.20. By what percentage do HipHop/Rap campaigns differ from Pop campaigns (the reference) — approximately and exactly?
2. A marketing manager proposes to double the promotion budget of every campaign. By roughly what percentage would streams increase, according to the model? Is this a causal claim?

Live demo: linear or log-log?

Course book, §5.7 — move the **true elasticity**.

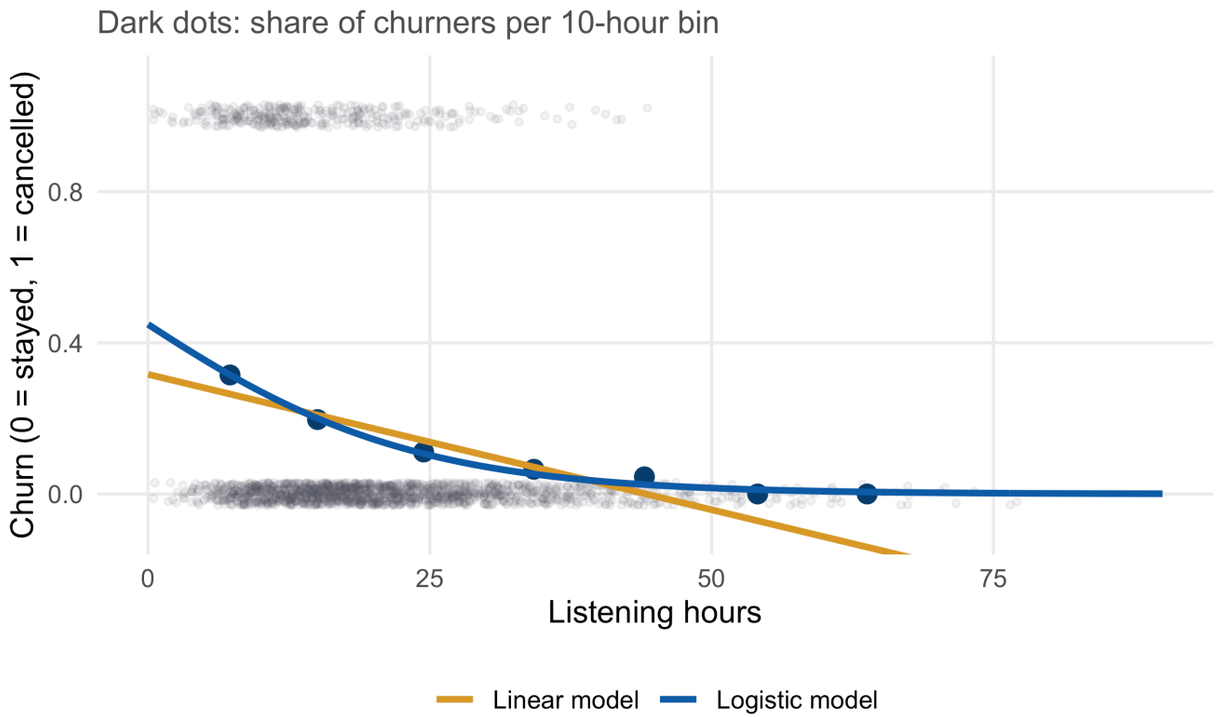
https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-form

Screenshot placeholder: the functional-form widget
`img/ch05/widget_loglog.png`

Elasticity **< 1**: diminishing returns, the line misses at both ends. **= 1**: proportional. **> 1**: increasing returns (rare — network effects).

5.8 Yes/no outcomes: logistic regression

Why not a straight line?



- Outcome: **churn** (cancelled within 3 months), 17% of users
- A line produces probabilities **below 0** for heavy listeners
- The **logistic** curve stays between 0 and 1 — and bends where it matters

```
model_churn <- glm(churn ~ hours + premium + tenure + smar
                   family = binomial, data = smartmix_obs)
```

Reading the coefficients: through probabilities

```r
model_churn <- glm(churn ~ hours + premium + tenure + smart_mix, family = binomial, data = smartmix_obs)
coef_table(model_churn)
```

```
            Estimate Std. Error z value p-value
(Intercept)    0.277      0.161    1.72   0.086
hours         −0.061      0.009   −7.13 < 0.001
premium       −0.533      0.153   −3.49 < 0.001
tenure        −0.020      0.004   −5.45 < 0.001
smart_mix     −0.256      0.162   −1.58   0.114
```

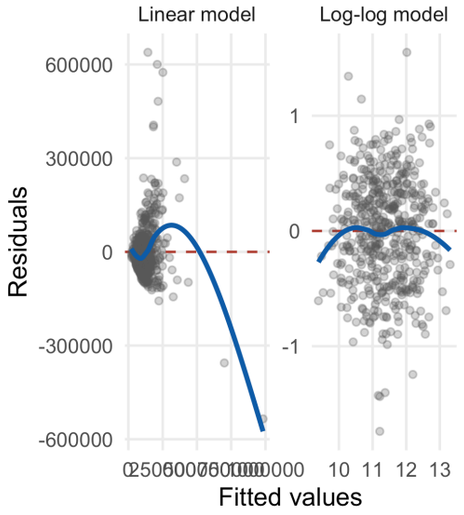
```r
typical_user <- data.frame(hours = c(10, 20, 30), premium = 0,
                           tenure = 24, smart_mix = 0)
predict(model_churn, newdata = typical_user, type = "response")
```

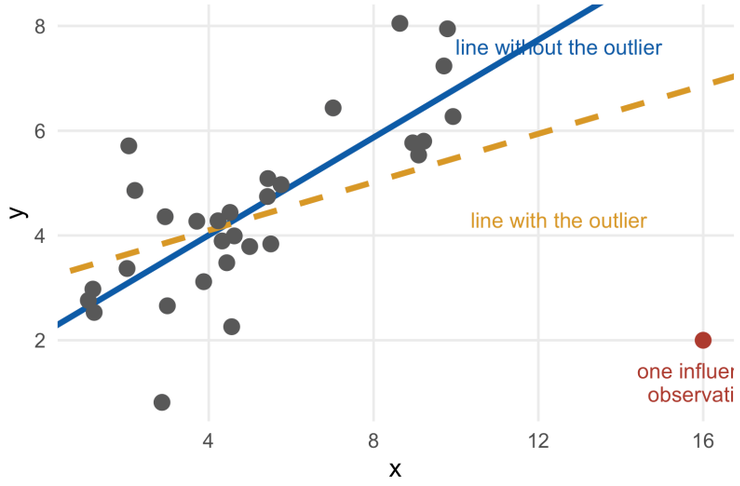
```
     1      2      3
0.3057 0.1936 0.1157
```

Read **sign** and **significance** directly; read **size** through **predicted probabilities**: 10 vs. 30 hours → churn probability 31% vs. 12%. Differences are effects in **percentage points**. Odds ratios: book box.

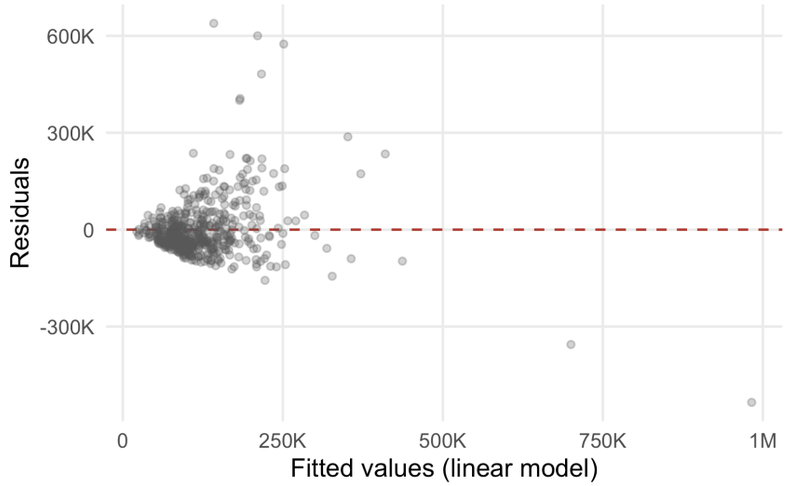
5.9 The diagnostics that change decisions

Three checks worth doing every time





The spread of the residuals grows with the fitted values — a fu



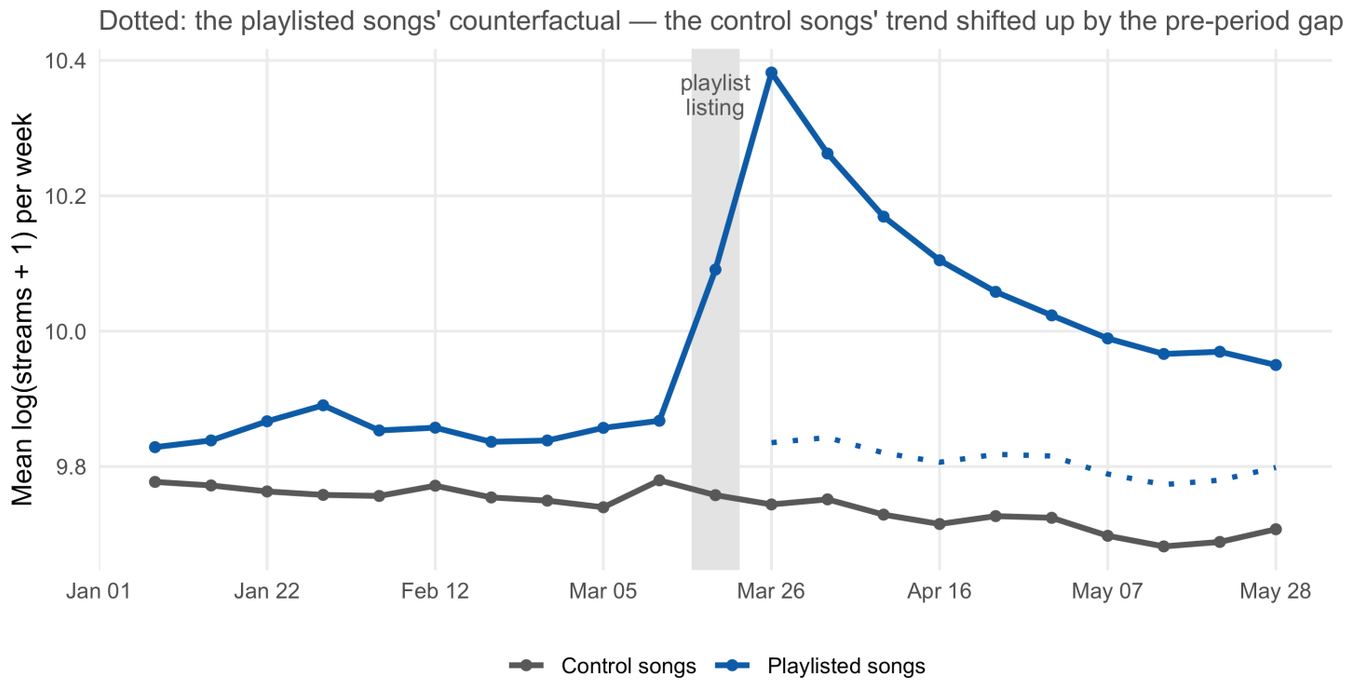
1. Is the line the right shape? Curved residuals → change the functional form (log) or add a predictor.

2. Does one observation drive the result? `plot(model, which = 5)`; refit without it — if the conclusion changes, say so.

3. Is the uncertainty right? A funnel → **robust standard errors**: `coeftest(model, vcov = vcovHC(model))`.

5.10 A quasi-experiment: difference-in-differences

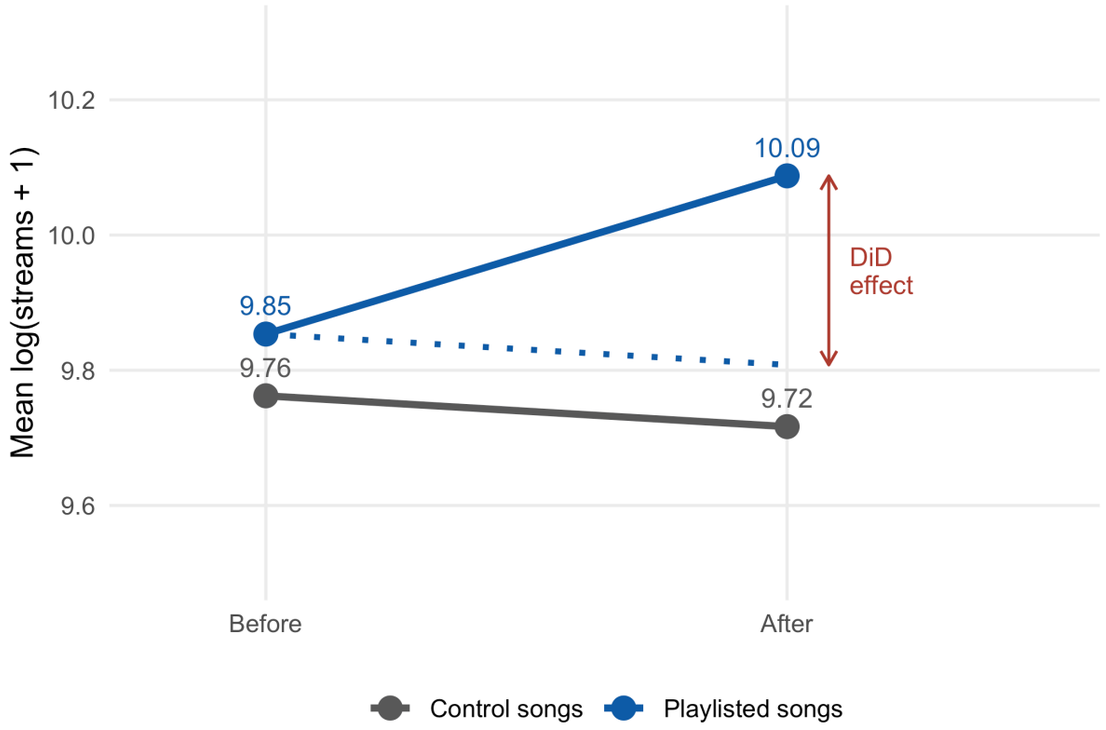
Playlist listings as a natural experiment



- **Real data:** 290 songs × 21 weeks
- March 19, 2018: **53 songs** added to a major playlist; 237 control songs
- Treated songs start **higher** → levels not comparable
- Before the listing: both groups move in **parallel**
- At the listing: the treated group **jumps**

Data: WU-RDS research project (github.com/WU-RDS/did).

Two differences, one estimate



$$\text{DiD} = (\bar{y}_{T,\text{after}} - \bar{y}_{T,\text{before}}) - (\bar{y}_{C,\text{after}} - \bar{y}_{C,\text{before}})$$

$(10.09 - 9.85) - (9.72 - 9.76) =$ **0.28** log points

> The control group's change estimates what would have happened to the treated songs **without** the listing — the counterfactual (dotted).

The DiD regression

```r
did_data <- playlist |>
  filter(week != as.Date("2018-03-19")) |>          # drop the listing week
  mutate(log_streams = log(streams + 1))
model_did <- lm(log_streams ~ treated * post, data = did_data)
coef_table(model_did)
```

```
            Estimate Std. Error t value p-value
(Intercept)    9.762      0.018  545.86 < 0.001
treated        0.091      0.042    2.18   0.029
post          −0.046      0.025   −1.80   0.071
treated:post   0.280      0.059    4.73 < 0.001
```

Coefficient	Meaning
`(Intercept)` 9.76	Control, before: baseline
`treated` 0.09	Treated, before: the **level difference**
`post` −0.05	Control, after − before: the **common time trend**
`treated:post` **0.28**	The extra change of the treated songs: the **DiD effect** → ≈ **+32% streams**

The assumption: parallel trends

```r
placebo_data <- did_data |>
  filter(post == 0) |>
  mutate(fake_post = ifelse(week >= as.Date("2018-02-12"), 1, 0))
coef_table(lm(log_streams ~ treated * fake_post,
              data = placebo_data))["treated:fake_post", ]
```

```
                Estimate Std. Error t value p-value
treated:fake_post    0.002      0.085    0.03   0.978
```

> Without the listing, the treated songs would have followed the **same trend** as the controls. **Untestable** — but the pre-period makes it plausible: a fake listing in the pre-period shows **no effect**.

Live demo: course book, §5.10

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-did

Your turn (5 min)

Add `major_label` as a control variable to the DiD regression:

```
coef_table(lm(log_streams ~ treated * post + major_label, data = did_data))
```

Does the effect change? Then: name one situation in which the parallel-trends assumption would **fail** for playlist listings.

What observational data can and cannot tell you

Two strategies, one honest report

- **Adjustment** handles confounders you can **measure** — only as good as your proxies; “enough” is untestable
- **Difference-in-differences** handles confounders **constant over time** — needs parallel trends, plausible but never proven
- **Prediction** needs no causal assumptions — a good predictor is valuable even if no coefficient is an effect

> Report effect + uncertainty + **assumptions**: “Adjusting for prior listening, tenure, Premium and device, adopters listen ≈ 4.5 hours more (95% CI 3.6–5.3), assuming no further differences affect listening — the experiment (3.2) suggests this is not fully met.”

Assignment 3: analyzing observational data

- Your **personal** data (based on your student ID): an opt-in feature, promotion campaigns, and a small panel with a listing
- Tasks: naive vs. adjusted · categorical predictor · elasticity · logistic regression · DiD
- **Deadline:** Group A Sun, Nov 22 · Group B Wed, Nov 25
- **Debrief** at the start of the following session

10 points: report 7 + debrief 3. State your **assumptions** — the best answers say what the data *cannot* show.

Learning check (Part II)

1. Log-log coefficient of ad spend = 0.15. Interpret. The level model says 0.8 sales per euro — why might the two disagree about doubling the budget?
2. Log-level model, dummy “promotion week” = 0.35. Percentage difference in sales — approximate and exact?
3. Residual plot shows a funnel. Consequence for coefficients, for *p*-values — and the simplest fix?
4. A DiD of a price promotion in treated stores: which assumption, and how can the pre-period help?

Next session

Module 6 · Mechanisms & heterogeneity — *Why, and for whom, does it work?*

- Interaction effects: when the effect differs between groups (moderation)
- Mediation: through which mechanism an effect works
- In experiments and observational data — and in your group project